

Can We Actually Look Inside an AI?

Mechanistic interpretability recap handout

Big idea

Mechanistic interpretability tries to move beyond surface-level stories about why a model answered a certain way. The goal is to make testable claims about what internal parts are doing and then check those claims by intervening inside the model.

The framework from the meeting

Step	What it means
1. Hypothesize	Propose what internal process might exist.
2. Localize	Figure out where that process seems to live.
3. Intervene	Patch, ablate, or steer part of the model.
4. Measure	Check whether the behavior changes in the predicted way.

Key ideas explained simply

- Surface explanation: a story told from the output alone. It can sound convincing even if it does not reflect the real internal process.
- Mechanistic explanation: a stronger claim about internal computation, supported by interventions and measurement.
- Polysemanticity: one neuron or unit can participate in several different concepts, so "one neuron = one idea" usually fails.
- Superposition: models may pack many sparse features into overlapping directions, which saves space but makes interpretation harder.

Why neurons are hard to read

The deck's polysemanticity example showed one unit participating in city names, bridges, quoted speech, praise, and code. That is useful for the model, but frustrating for humans trying to assign one clean meaning to one neuron.

Why features overlap

The superposition section used the idea of several features sharing fewer clean axes. If only a few features are active at once, overlap can still work, but it becomes much harder to point to one coordinate and say exactly what it means.

Main methods from the lesson

Activation patching was presented as a causal swap test: run a clean prompt and a corrupted prompt, replace one hidden state in the bad run with the matching state from the clean run, and see whether behavior recovers.

IOI demo (Indirect Object Identification): the model must continue a sentence with the correct person. The deck used the contrast between "Mary gave ... to __" and "John gave ... to __" to show how one changed token can reveal where useful internal information lives.

Sparse autoencoders (SAEs): these convert dense, mixed activations into a sparse code with fewer active features. The meeting stressed that SAEs do not reveal the full computation, but they often produce more usable units than raw neurons.

Feature visualization and steering: a good feature should light up across related contexts, not just one exact phrase. The deck used a "Golden Gate Bridge" feature across languages and modalities, then showed that increasing a feature can steer outputs in a predictable direction.

Method	What it can show	Main limitation
Activation patching	Whether a specific hidden state causally matters for behavior.	Results depend on corruption setup, metric, and patch location.
SAEs	Cleaner candidate features than raw neurons.	Features can still be incomplete or only part of the computation.
Feature steering	Whether increasing or decreasing a feature changes output predictably.	A steering effect is evidence, not a complete explanation.

Balanced takeaway

- These tools give real evidence about model internals. They are better than pure guesswork from outputs alone.
- They still do not produce full transparency for large frontier models.
- The right attitude is neither hype nor cynicism: partial understanding is still useful.

One-sentence summary

Mechanistic interpretability is the effort to turn AI understanding into a science of hypotheses, interventions, and tested mechanisms.

Discussion questions from the meeting

- Would you trust AI more if researchers could inspect its internal circuits better?
- Is checking outputs enough, or do we also need to understand the inside?
- If we can only partly understand a model, is that still useful?

Prepared from the AI Club slide deck for recap use.